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Early-onset cancer incidence in the United States by race/ethnicity between 2011 and 2020

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ABSTRACT

We characterized trends in early onset (aged 20–49) cancer incidence by race/ethnicity and sex using the 2011–2020 Surveillance, Epidemiology, and End Results (SEER) Program dataset. We estimated age-standardized cancer incidence rates, incidence rate ratios (IRR), and annual percentage changes (APC) with 95 % confidence intervals (CI). During the time period examined, cancer incidence increased for female breast (APC: 0.64; 95 % CI: 0.10, 1.20), female colorectal (APC: 2.16; 95 % CI: 1.22, 3.10), and male colorectal (APC: 2.49; 95 % CI: 1.81, 3.19) cancer. Among racial/ethnic groups examined, Hispanic individuals had the largest increases in female all sites (APC: 1.31; 95 % CI: 0.38, 2.25), female breast (APC: 1.04; 95 % CI: 0.29, 1.81), and female (APC: 4.67; 95 % CI: 3.07, 6.30) and male (APC: 3.53; 95 % CI: 2.58, 4.49) colorectal cancer incidence. Further research is needed to clarify the causal mechanisms driving these patterns.

1. Brief communication

In 2024, over 2 million new cancer cases and 600,000 cancer deaths are projected to occur in the United States [1]. While substantial progress across the cancer care continuum has led to significant declines in cancer mortality rates in the last few decades, cancer incidence rates have increased among younger adults [2]. In a study of data from 2010 to 2019, breast cancer had the highest number of incident cases, with gastrointestinal cancers having the fastest growing incidence rates among all early-onset cancers [3]. We build on prior research evaluating increases in early onset cancer (aged 20–49) incidence by describing differences in these trends by race/ethnicity and sex.

In this cross-sectional study, we used 2011–2020 data from the Surveillance, Epidemiology, and End Results (SEER) Program [4]. The SEER Program provides a comprehensive source of population-based information in the United States that includes patient demographics, primary tumor site, tumor morphology, stage at diagnosis, first course of treatment, and follow-up for vital status. SEER incidence data is obtained from population-based registries covering approximately 48 % of the US population [5]. We used publicly available deidentified data, so institutional review board approval and informed consent were not necessary. This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines for cross-sectional studies.

SEER*Stat software, version 8.4.3, was used to estimate age-standardized cancer incidence rates with 95 % confidence intervals (95 % CI) for cancer diagnosed in 2011–2020. All incidence rates were age-standardized to the 2000 United States population. Incidence rate ratios (IRR) with 95 % CI were calculated for the population aged 50 + vs. 20–49 for all cancers and the 10 most common cancers. For individuals in each age group, incidence rate ratios were further calculated for race/ethnicity (Non-Hispanic [NH]-White, NH-Black, Hispanic) and sex (male, female). Moreover, we calculated annual percent changes (APC) and 95 % CI for the overall period from 2011 to 2020 for the four most common cancers (female breast, prostate, lung and bronchus, and colorectal) [6] by age group (50 +, 20–49) and by race/ethnicity (NH-White, NH-Black, Hispanic) among patients aged 20–49. Given disruptions in cancer screening and diagnosis due to the COVID-19 pandemic in 2020 [7], we also conducted a sensitivity analyses calculating APCs and 95 % CIs by age group for cancer diagnosed in 2011–2019.

Between 2011 and 2020, cancer incidence rates decreased for individuals aged 50 + for all cancer types evaluated, with the greatest decreases observed for male lung and bronchus (APC: − 3.24; 95 % CI:

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− 4.05, − 2.43), female colorectal (APC: − 2.47; 95 % CI: − 3.24, −1.181), and male colorectal (APC: − 2.56; 95 % CI: − 3.44, − 1.67) (Fig. 1). However, for individuals aged 20–49, cancer incidence rates increased for female breast (APC: 0.64; 95 % CI: 0.10, 1.20), female colorectal (APC: 2.16; 95 % CI: 1.22, 3.10), and male colorectal (APC: 2.49; 95 % CI: 1.81, 3.19). Results from sensitivity analyses limited to data from 2011–2019 were largely consistent, with some suggestion that cancer incidence rates have also increased for female all sites (APC: 0.66; 95 % CI: 0.35, 0.97) (Supplemental Table 1).

Cancer IRRs and 95 % CI by age group, race/ethnicity, and sex are presented in Supplemental Table 2. Among individuals aged 20–49, females had a 70 % higher incidence rate for all cancers overall compared with males (IRR: 1.70; 95 % CI: 1.70, 1.71), NH-Black individuals had almost 3 times the incidence of prostate cancer compared with NH-White individuals (IRR: 2.69; 95 % CI: 2.60–2.77), while Hispanic (IRR: 0.43; 95 % CI: 0.41, 0.45) and NH-Black (IRR: 0.49; 95 % CI: 0.45, 0.52) individuals had approximately half the incidence of bladder cancer compared with NH-White individuals.

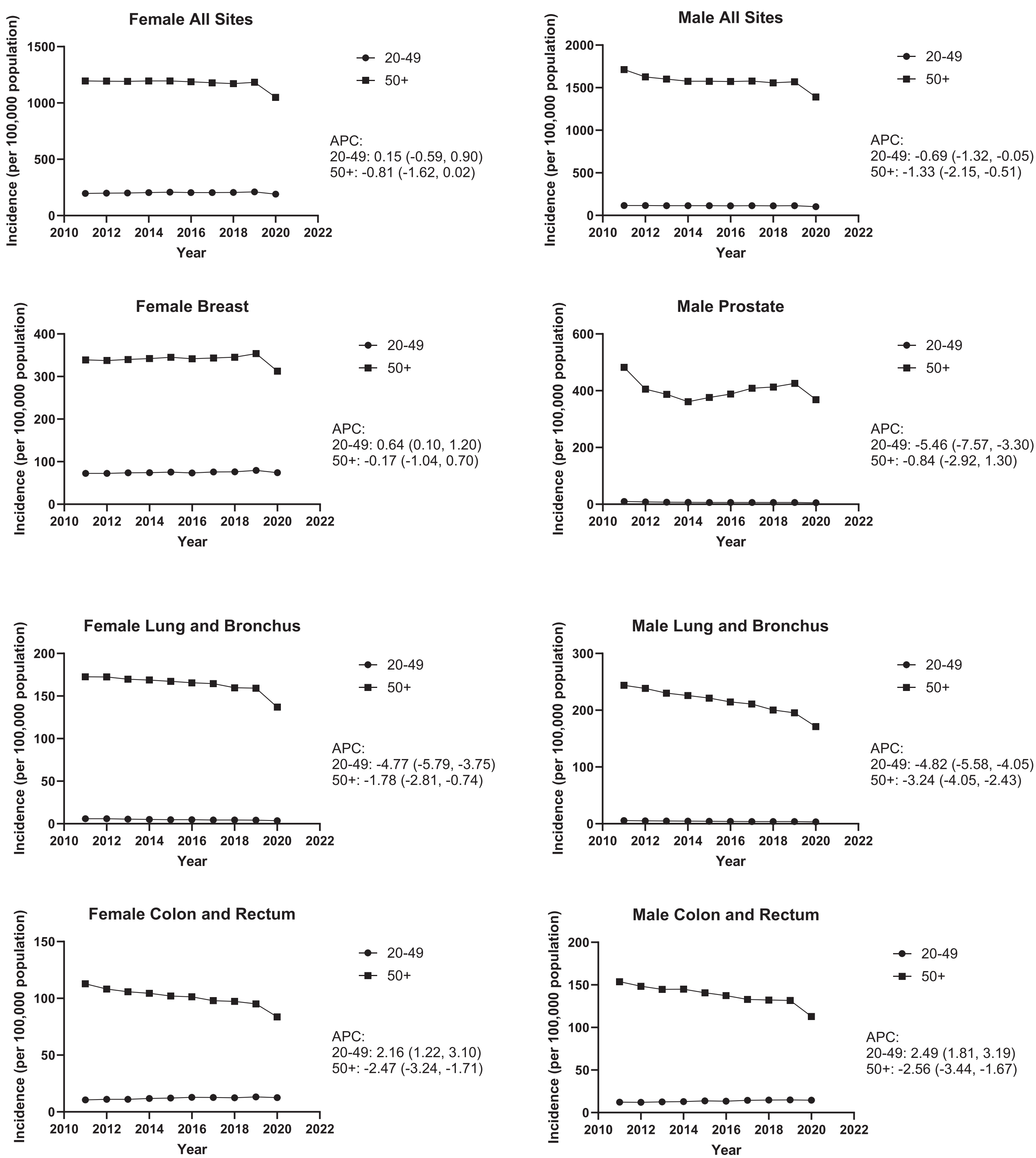

Fig. 1. Age-standardized cancer incidence rates and 95 % confidence intervals by age group, 2011–2020. APC = annual percent change.

Longitudinal trends among individuals aged 20–49 are presented in Fig. 2 by race/ethnicity. Decreases were observed for male prostate, female lung and bronchus, and male lung and bronchus across all racial/ethnic groups evaluated. Despite having the lowest incidence rates of the racial/ethnic groups examined, Hispanic individuals had the largest increases in female all sites (APC: 1.31; 95 % CI: 0.38, 2.25), female breast (APC: 1.04; 95 % CI: 0.29, 1.81), and female (APC: 4.67; 95 % Cl: 3.07, 6.30) and male (APC: 3.53; 95 % CI: 2.58, 4.49) colorectal cancer incidence during the study period.

Although age-standardized cancer incidence rates have declined significantly between 2011 and 2020 in the United States among individuals aged 50+ for all major cancer types, we found that incidence rates have risen among individuals aged 20–49 for the following subgroups: female breast, female colorectal, and male colorectal. Among the racial/ethnic groups examined, the largest percent increases for breast and colorectal cancer were observed for Hispanic individuals

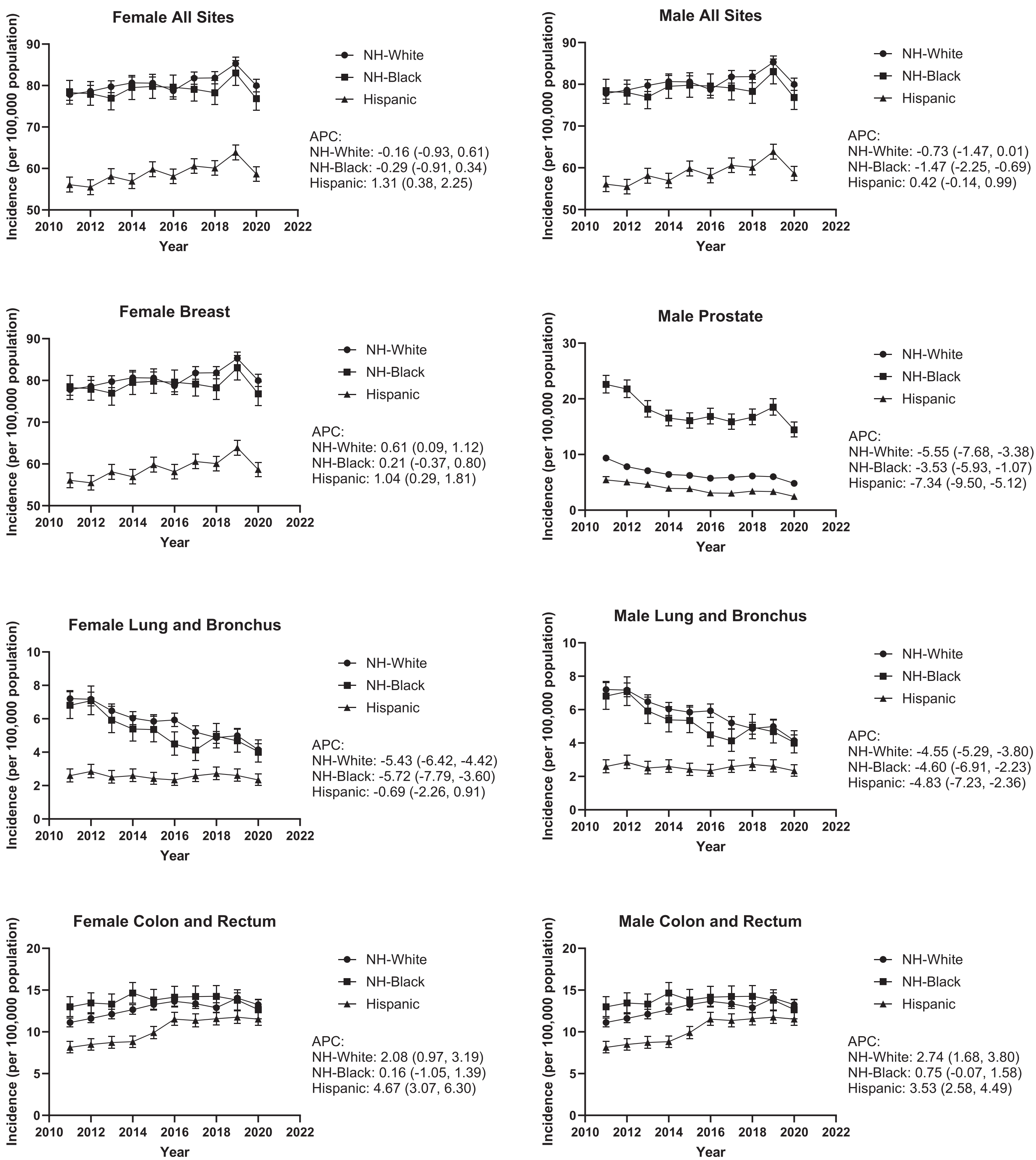

Fig. 2. Age-standardized cancer incidence rates and 95 % confidence intervals by race/ethnicity among adults aged 20–49, 2011–2020. APC = annual percent change; NH = non-Hispanic.

compared to NH-White and NH-Black individuals. Increases in early onset cancer have been attributed to rising obesity [2,8], exposure to radiation, earlier age at menarche, and changes in the microbiome [9]. There is also increasing evidence highlighting the role of early childhood factors from the early prenatal to adolescent periods of life, which may cause genetic and epigenetic alterations that promote carcinogenesis and affect immunity [10,11]. These risk factors may be unequally distributed among different population groups in the United States, resulting in the observed patterns. Indeed, prior research has documented that those who are Hispanic may be disproportionately affected by increases in early-onset cancer [3,12]. However, further research is needed to clarify the causal mechanisms driving these increases.

There are several limitations that should be considered in the interpretation of our results. First, there were massive disruptions in cancer screening and diagnosis due to the COVID-19 pandemic, leading to underreporting and apparent declines in 2020 incidence rates [7]. Second, there is the potential for misclassification of race/ethnicity and cause of death; however, this is unlikely to be differential by population subgroups. Third, we acknowledge that the SEER incidence data covers less than 50 % of the US population [5], and thus our results may not be fully generalizable to the entire country. Still, important strengths of this analysis include the large sample size, enabling us to identify subgroups experiencing the most significant increases in early-onset cancer incidence.

In conclusion, we found that cancer incidence rates have risen among young adults over the last decade, particularly for breast and colorectal cancer. Future research is needed to identify the causal mechanisms driving increases in early onset cancer incidence.

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Disclosures

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Anjali Gupta: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Tomi Akinyemiju:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors have no disclosures.

Data Availability

This analysis utilized secondary data from the SEER Program. More information and instructions to request these data are available here: https://seer.cancer.gov/data/access.html.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.canep.2024.102632.

References

[1] R.L. Siegel, A.N. Giaquinto, A. Jemal, Cancer statistics, 2024, CA Cancer J. Clin. 74 (1) (2024) 12–49.

[2] A. Bleyer, Increasing cancer in adolescents and young adults: cancer types and causation implications, J. Adolesc. Young Adult Oncol. 12 (3) (2023) 285–296.

[3] B. Koh, D.J.H. Tan, C.H. Ng, et al., Patterns in cancer incidence among people younger than 50 years in the US, 2010–2019, JAMA Netw. Open 6 (8) (2023) e2328171.

[4] Surveillance, Epidemiology, and End Results (SEER) Program (⟨www.seer.cancer.gov⟩) SEER*Stat Database: Incidence - SEER Research Plus Limited-Field Data, 22 Registries, Nov 2022 Sub (2000–2020) - Linked To County Attributes - Total U.S., 1969–2021 Counties, National Cancer Institute, DCCPS, Surveillance Research Program, released April 2023, based on the November 2022 submission.

[5] SEER, Information About Cancer Statistics, National Cancer Institute. ⟨https://seer.cancer.gov/about/using-website.html#:~:text=SEER%20collects%20and%20publishes%20cancer,from%2022%20U.S.%20geographic%20areas⟩. (Accessed 8 February 2024).

[6] R.L. Siegel, K.D. Miller, N.S. Wagle, A. Jemal, Cancer statistics, 2023, CA: A Cancer J. Clin. 73 (1) (2023) 17–48.

[7] A.B. Mariotto, E.J. Feuer, N. Howlader, H.S. Chen, S. Negoita, K.A. Cronin, Interpreting cancer incidence trends: challenges due to the COVID-19 pandemic, J. Natl. Cancer Inst. 115 (9) (2023) 1109–1111.

[8] H. Li, D. Boakye, X. Chen, et al., Associations of body mass index at different ages with early-onset colorectal cancer, Gastroenterology 162 (4) (2022) 1088–1097, e1083.

[9] T. Ugai, N. Sasamoto, H.Y. Lee, et al., Is early-onset cancer an emerging global epidemic? Current evidence and future implications, Nat. Rev. Clin. Oncol. 19 (10) (2022) 656–673.

[10] N. Akimoto, T. Ugai, R. Zhong, et al., Rising incidence of early-onset colorectal cancer – a call to action, Nat. Rev. Clin. Oncol. 18 (4) (2021) 230–243.

[11] A.M. Bever, M. Song, Early-life exposures and adulthood cancer risk: a life course perspective, J. Natl. Cancer Inst. 115 (1) (2023) 4–7.

[12] C. Muller, E. Ihionkhan, E.M. Stoffel, S.S. Kupfer, Disparities in early-onset colorectal cancer, Cells 10 (5) (2021).